

Caijun Qin

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Google Scholar: <https://scholar.google.com/citations?hl=en&user=aRGtWp0AAAAJ>

EDUCATION

Jan 2025 – Dec 2026 <i>Expected</i>	GEORGIA INSTITUTE OF TECHNOLOGY ♦ Atlanta, GA, USA Degree: Computer Science M.S. Courses: Software Analysis, Network Science, Advanced Operating Systems, Deterministic Optimization
Aug 2019 – May 2022	UNIVERSITY OF FLORIDA ♦ Gainesville, FL, USA Major: Computer Science B.S. & Statistics B.A. Major GPA: 3.79 / 4.00 Courses: Software Engineering, Data Structures and Algorithms, Machine Learning, Performant Programming in Python, Competitive Programming, Principles of Programming Language, Data Science, Programming with Data in R, Statistical Learning
Aug 2018 – May 2019	FLORIDA STATE UNIVERSITY ♦ Tallahassee, FL, USA Major: Computer Science B.S. Major GPA: 4.00 / 4.00 Courses: Intro Programming I, Object-oriented Programming, Linear Algebra, Calculus III

RESEARCH INTERESTS

Graph Theory, Network Algorithms, Programming Language Design, Compilers, Discrete Math

RESEARCH EXPERIENCE

Jan 2022 – Present	Lake Kivu Plant Pathology Analysis ♦ Undergraduate Research, Dept. of Plant Pathology, IFAS, UF <i>Undergraduate Research Assistant ♦ Department of Plant Pathology, IFAS, University of Florida</i> <i>Advisor: Dr. Karen Garrett</i> - Applied NASA Power API within prediction models to identify possible crop disease by location data, leveraging H2O AutoML to reduce total time to train, evaluate, and rank models by ~10-fold - Standardized data cleaning with pipelines built using Apache Spark extensions in R
Jun 2021 – Present	Using AI to Trace the History of Race and Inequality <i>Undergraduate Research Assistant ♦ Department of Classics, College of Liberal Arts & Sciences, University of Florida</i> <i>Advisor: Dr. Eleni Bozia</i> - Engineered NLP pipeline to retrieve, transform, and index Latin and Greek from archived XML sources, allowing easy search to support sentiment analysis of daily life in the classical world - Supervised undergraduate teams on digital text extraction and XML indexing to compare metadata between pieces of text volumes during text processing
Jun 2021 – Aug 2021	“Data Analytics and Information Retrieval” NSF Research Experience for Undergraduates <i>NSF REU Recipient ♦ Department of Information Science, College of Information, University of North Texas</i> <i>Advisor: Dr. Junhua Ding</i> - Experimentally evaluated traditional versus deep learning algorithms on legal text classification to identify models that could help legal teams quickly identify legal contract type from corporate documents - Analyzed dataset factors to explain why traditional ML, especially boosting algorithms, outperformed deep learning models transferred from BERT and Sentence-BERT models
May 2020 – April 2021	Predictive Sampling Method for Spread Models in Networks <i>Undergraduate Researcher ♦ University Scholars program, University of Florida</i> <i>Advisor: Dr. Peter Dobbins</i> - Developed a novel, trainable sampling method for large networks by targeting high-degree nodes in a network to test the addition of parameter tuning that is largely lacking in existing sampling algorithms - Applied statistical measurements to compare with existing methods to crawl networks, which hinted community detection applications for the new algorithm particularly in dense networks

PROJECT EXPERIENCE

- Feb 2022 – Apr 2022 **Evaluating Tree-based Models for Survival Analysis ♦**
Course Project ♦ Department of Statistics, University of Florida
- Compared selection of tree-based machine learning methods against traditional survival analysis methods, namely Kaplan-Meier and Cox-Hazard, on predicting lifetimes of Kickstarter projects being funded
 - Experimental results concluded the traditional methods performed similarly to each other but were dominated by tree-based methods, most prominently conditional inference trees across all scoring metrics
- Aug 2021 – Dec 2021 **Plant Root Analysis Using Machine Learning**
Collaborator ♦ Senior Project Course, University of Florida
Advisor: Dr. Alina Zare
- Performed image segmentation of minirhizotron (MR) root images using U-Net deep learning architecture
 - Improved original model by tweaking hyperparameters and training with only subset of original training data
 - Programmed framework to apply multiple cumulative learning paradigms on models created with PyTorch
- May 2020 – Aug 2020 **OCR Note-taking Application**
Backend Engineer ♦ Performant Programming Course, University of Florida
- Integrated machine learning and database functionalities for optical character recognition (OCR) app
 - Maintained customizable pipeline to preprocess text images with OpenCV and Pillow
 - Experimentally optimized parameter selection and preprocessing steps to boost text prediction accuracy
- Jan 2019 – May 2019 **Cost Minimization and Optimization of Criteria-based Matchings**
Student Researcher ♦ University Honors Program, Florida State University
Advisor: Dr. Peixiang Zhao
- Created a CLI program for assignment optimization in operations research, leveraging the Hungarian algorithm to minimize total cost in matching worker to tasks
 - Integrated additional variant algorithm based on maximizing sum of desirable attribute values
 - Designed similar algorithm but which optimally chooses pairs based on sum of positive attribute values

PROFESSIONAL EXPERIENCE

- Oct 2022 – Jan 2025 **Software Development Engineer**
Amazon Web Services, Seattle, WA
- Created distributed agent system to upkeep 240+ EC2 instances across 40+ regions under OS patching campaign, providing an automated solution to meet security SLAs
 - Developed dashboard plot feature for diagnostics website to enhance visibility in NLB network traffic, resulting in faster access to metrics for troubleshooting
 - Automated logic to bulk refresh ALB/CLB API workflows as batch jobs of up to 500 workflows at a time, leading to large time savings for engineers when retrying failed workflows on diagnostics website
 - Delivered infrastructure build for canary testing service and diagnostics website to 9 new regions, ensuring new AWS regions receive testing of different load balancer services on schedule

PUBLICATIONS

- Qin, C.,** Yang, Y., Chen, H., & Ding, J. (2021). *A Comparison Study of Machine Learning and Deep Learning for Legal Contract Understanding* [Manuscript submitted for publication], Department of Computer & Information Science & Engineering (CISE), University of Florida.
- Qin, C.** (2021). Predictive Sampling Method for Spread Models in Networks. *UF Journal of Undergraduate Research*, 23(Fall 2021). <https://doi.org/10.32473/ufjur.v23i.128429>

PRESENTATIONS

- Qin, C.,** Yang, Y., Chen, H., Ding, J. (2021, November). A Comparison Study of Machine Learning and Deep Learning for Legal Contract Understanding [Paper presentation]. In *International Workshop on Juris-Informatics 2021 (JURISIN 2021)* (pp. 110–123), Keio University, Yokohama, Kanagawa, Japan.
- Qin, C.** (2021, March). Predictive Sampling Method for Spread Models in Networks. In *2021 Virtual Spring Undergraduate Research Symposium*, University of Florida, Gainesville, Florida, USA.

AWARDS & HONORS

Jan 2021	InfoTech Challenge for most innovative use of a public dataset for the public good <i>SwampHacks VII ♦ University of Florida, Gainesville, FL, USA</i>
Mar 2019	1st Place in Lower Division <i>FSU Spring 2019 Programming Competition ♦ Florida State University, Tallahassee, FL, USA</i>
Jan 2019	University Honors Program <i>Lateral Admission into Honors Program ♦ Florida State University, Tallahassee, FL, USA</i>

FUNDING & SCHOLARSHIPS

Jun 2021	NSF REU: College of Information at UNT <i>\$7000 ♦ University of North Texas, Denton, TX, USA</i>
Mar 2021	Gartner Group Information Technology Fund <i>\$1000 ♦ University of Florida, Gainesville, FL, USA</i>
May 2020	Russell and Mary Hyatt McCaughan Scholarship <i>\$1000 ♦ University of Florida, Gainesville, FL, USA</i>
Feb 2020	University Scholars Program Stipend <i>\$1750 ♦ University of Florida, Gainesville, FL, USA</i>
May 2018	University Freshman Scholarship <i>\$1200 / Semester ♦ Florida State University, Tallahassee, FL, USA</i>

CERTIFICATIONS

Mar 2022	NVIDIA Deep Learning Institute: Building Transformer-Based NLP Applications
Feb 2022	NVIDIA DLI: Applications of AI for Anomaly Detection
Apr 2021	MATLAB Machine Learning Onramp
Apr 2021	MATLAB Onramp
Jul 2020	M001: MongoDB Basics
May 2015	Microsoft Office Specialist: Word 2013
May 2015	Microsoft Office Specialist: Excel 2013
May 2015	Microsoft Office Specialist: PowerPoint 2013

LEADERSHIP & SERVICE

Nov 2021 –	Data Science and Informatics (DSI)
May 2022	<i>Project Coordinator ♦ University of Florida</i>
Aug 2021 –	UF Chapter of Association of Computing Machinery (ACM)
May 2022	<i>Treasurer ♦ University of Florida</i>
Aug 2021 –	Open Source Club (OSC)
May 2022	<i>Project Lead ♦ University of Florida</i>
Aug 2021 –	Google Developer Student Club (GDSC)
May 2022	<i>Tech Lead ♦ University of Florida</i>
Aug 2021 –	Software Engineering Club (SEC)
May 2022	<i>Webmaster ♦ University of Florida</i>
Apr 2019 –	FSU Chapter of Association of Computing Machinery (ACM)
Aug 2019	<i>Treasurer ♦ Florida State University</i>

PROFESSIONAL SKILLS

Programming Languages	Python, C/C++, Java, R, Julia, Ruby, JavaScript, TypeScript, OCaml
Markup Languages	JSON, HTML, CSS, XML, YAML
Databases	SQLite, MySQL, MongoDB, PostgreSQL
Frameworks & Other Tools	AWS services, Bash scripting, Linux utilities, Docker, Kubernetes, Apache Spark, LLVM, OpenMP, MPI, Node.js